

Thermodynamics

Instructor

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Chapter 11 – Problems

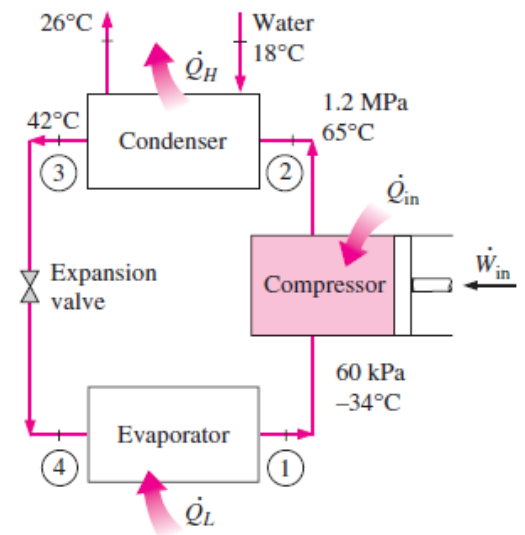
**Problem numbers correspond to
5th Edition**

The Reversed Carnot Cycle

11–2 A steady-flow Carnot refrigeration cycle uses refrigerant-134a as the working fluid. The refrigerant changes from saturated vapor to saturated liquid at 30°C in the condenser as it rejects heat. The evaporator pressure is 160 kPa. Show the cycle on a T - s diagram relative to saturation lines, and determine (a) the coefficient of performance, (b) the amount of heat absorbed from the refrigerated space, and (c) the net work input. *Answers: (a) 5.64, (b) 147 kJ/kg, (c) 26.1 kJ/kg*

Ideal and Actual Vapor-Compression Refrigeration Cycles

11-11 A commercial refrigerator with refrigerant-134a as the working fluid is used to keep the refrigerated space at -30°C by rejecting its waste heat to cooling water that enters the condenser at 18°C at a rate of 0.25 kg/s and leaves at 26°C . The refrigerant enters the condenser at 1.2 MPa and 65°C and leaves at 42°C . The inlet state of the compressor is 60 kPa and -34°C and the compressor is estimated to gain a net heat of 450 W from the surroundings. Determine (a) the quality of the refrigerant at the evaporator inlet, (b) the refrigeration load, (c) the COP of the refrigerator, and (d) the theoretical maximum refrigeration load for the same power input to the compressor.



Analysis (a) From refrigerant-134a tables (Tables A-11 through A-13)

$$\left. \begin{array}{l} P_1 = 60\text{ kPa} \\ T_1 = -34^{\circ}\text{C} \end{array} \right\} h_1 = 230.03\text{ kJ/kg}$$

$$\left. \begin{array}{l} P_2 = 1200\text{ kPa} \\ T_2 = 65^{\circ}\text{C} \end{array} \right\} h_2 = 295.16\text{ kJ/kg}$$

$$\left. \begin{array}{l} P_3 = 1200\text{ kPa} \\ T_3 = 42^{\circ}\text{C} \end{array} \right\} h_3 = 111.23\text{ kJ/kg}$$

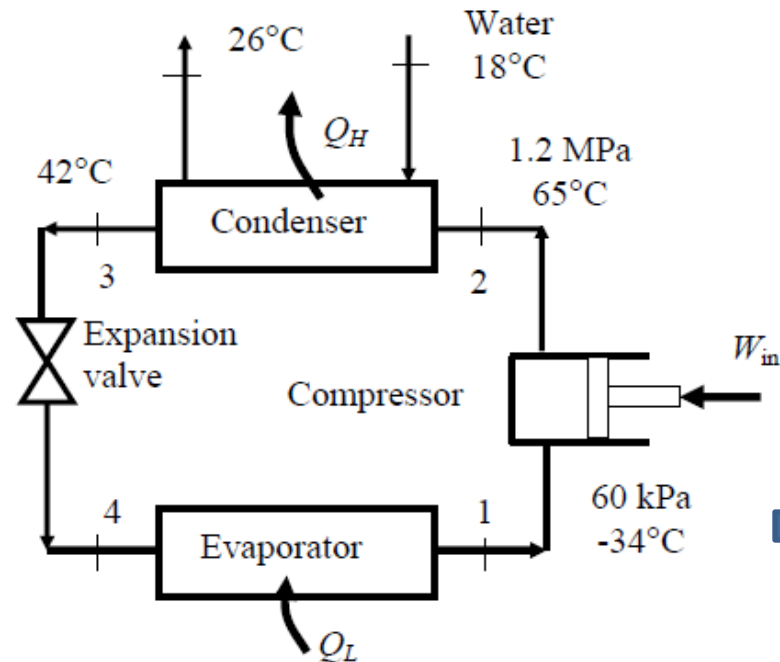
$$h_4 = h_3 = 111.23\text{ kJ/kg}$$

$$\left. \begin{array}{l} P_4 = 60\text{ kPa} \\ h_4 = 111.23\text{ kJ/kg} \end{array} \right\} x_4 = \mathbf{0.4795}$$

Using saturated liquid enthalpy at the given temperature, for water we have (Table A-4)

$$h_{w1} = h_f @ 18^{\circ}\text{C} = 75.47\text{ kJ/kg}$$

$$h_{w2} = h_f @ 26^{\circ}\text{C} = 108.94\text{ kJ/kg}$$



(b) The mass flow rate of the refrigerant may be determined from an energy balance on the compressor

$$\begin{aligned}\dot{m}_R(h_2 - h_3) &= \dot{m}_w(h_{w2} - h_{w1}) \\ \dot{m}_R(295.16 - 111.23)\text{kJ/kg} &= (0.25\text{ kg/s})(108.94 - 75.47)\text{kJ/kg} \\ \longrightarrow \dot{m}_R &= 0.0455\text{ kg/s}\end{aligned}$$

The waste heat transferred from the refrigerant, the compressor power input, and the refrigeration load are

$$\begin{aligned}\dot{Q}_H &= \dot{m}_R(h_2 - h_3) = (0.0455\text{ kg/s})(295.16 - 111.23)\text{kJ/kg} = 8.367\text{ kW} \\ \dot{W}_{\text{in}} &= \dot{m}_R(h_2 - h_1) - \dot{Q}_{\text{in}} = (0.0455\text{ kg/s})(295.16 - 230.03)\text{kJ/kg} - 0.45\text{ kW} = 2.513\text{ kW} \\ \dot{Q}_L &= \dot{Q}_H - \dot{W}_{\text{in}} = 8.367 - 2.513 = \mathbf{5.85\text{ kW}}\end{aligned}$$

(c) The COP of the refrigerator is determined from its definition

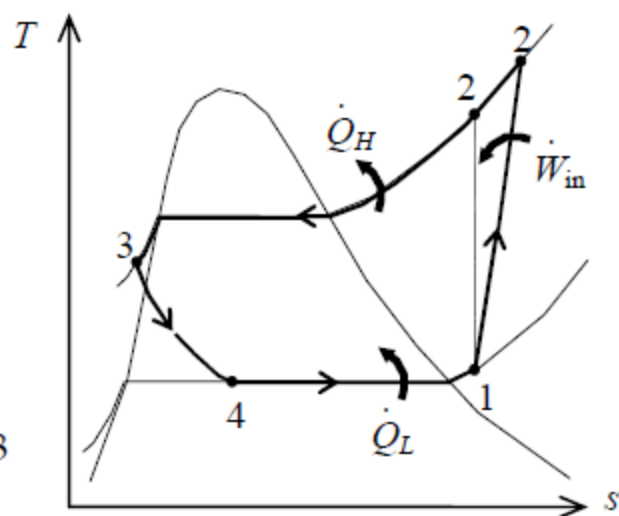
$$\text{COP} = \frac{\dot{Q}_L}{\dot{W}_{\text{in}}} = \frac{5.85}{2.513} = \mathbf{2.33}$$

(d) The reversible COP of the refrigerator for the same temperature limits is

$$\text{COP}_{\text{max}} = \frac{1}{T_H / T_L - 1} = \frac{1}{(18 + 273) / (-30 + 273) - 1} = 5.063$$

Then, the maximum refrigeration load becomes

$$\dot{Q}_{L,\text{max}} = \text{COP}_{\text{max}} \dot{W}_{\text{in}} = (5.063)(2.513\text{ kW}) = \mathbf{12.72\text{ kW}}$$



11-12 A refrigerator uses refrigerant-134a as the working fluid and operates on an ideal vapor-compression refrigeration cycle between 0.12 and 0.7 MPa. The mass flow rate of the refrigerant is 0.05 kg/s. Show the cycle on a T - s diagram with respect to saturation lines. Determine (a) the rate of heat removal from the refrigerated space and the power input to the compressor, (b) the rate of heat rejection to the environment, and (c) the coefficient of performance. *Answers: (a) 7.41 kW, 1.83 kW, (b) 9.23 kW, (c) 4.06*

11-13 Repeat Prob. 11-12 for a condenser pressure of 0.9 MPa.

11-14 If the throttling valve in Prob. 11-12 is replaced by an isentropic turbine, determine the percentage increase in the COP and in the rate of heat removal from the refrigerated space. *Answers: 4.2 percent, 4.2 percent*

11-18 Refrigerant-134a enters the compressor of a refrigerator as superheated vapor at 0.14 MPa and -10°C at a rate of 0.12 kg/s, and it leaves at 0.7 MPa and 50°C . The refrigerant is cooled in the condenser to 24°C and 0.65 MPa, and it is throttled to 0.15 MPa. Disregarding any heat transfer and pressure drops in the connecting lines between the components, show the cycle on a T - s diagram with respect to saturation lines, and determine (a) the rate of heat removal from the refrigerated space and the power input to the compressor, (b) the isentropic efficiency of the compressor, and (c) the COP of the refrigerator. *Answers: (a) 19.4 kW, 5.06 kW, (b) 82.5 percent, (c) 3.83*

$$\left. \begin{array}{l} P_1 = 0.14 \text{ MPa} \\ T_1 = -10^{\circ}\text{C} \end{array} \right\} \begin{array}{l} h_1 = 246.36 \text{ kJ/kg} \\ s_1 = 0.97236 \text{ kJ/kg} \cdot \text{K} \end{array}$$

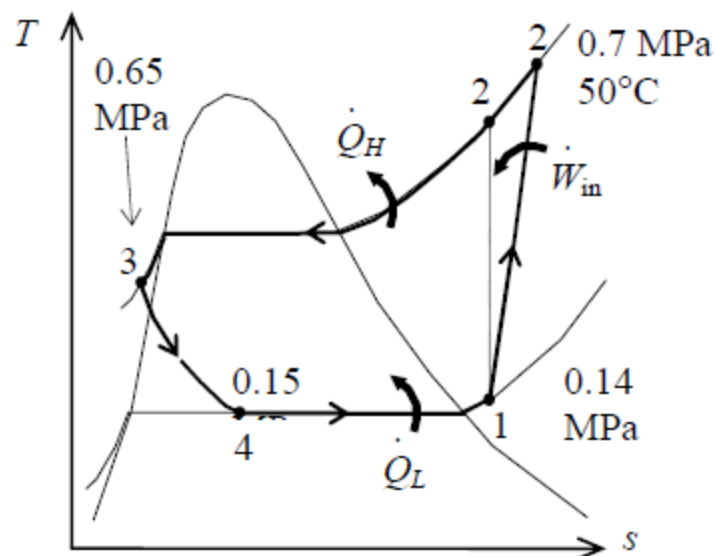
$$\left. \begin{array}{l} P_2 = 0.7 \text{ MPa} \\ T_2 = 50^{\circ}\text{C} \end{array} \right\} h_2 = 288.53 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_{2s} = 0.7 \text{ MPa} \\ s_{2s} = s_1 \end{array} \right\} h_{2s} = 281.16 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_3 = 0.65 \text{ MPa} \\ T_3 = 24^{\circ}\text{C} \end{array} \right\} h_3 = h_f @ 24^{\circ}\text{C} = 84.98 \text{ kJ/kg}$$

$$h_4 \cong h_3 = 84.98 \text{ kJ/kg (throttling)}$$

Then the rate of heat removal from the refrigerated space and the power input to the compressor are determined from



$$\dot{Q}_L = \dot{m}(h_1 - h_4) = (0.12 \text{ kg/s})(246.36 - 84.98) \text{ kJ/kg} = \mathbf{19.4 \text{ kW}}$$

and

$$\dot{W}_{\text{in}} = \dot{m}(h_2 - h_1) = (0.12 \text{ kg/s})(288.53 - 246.36) \text{ kJ/kg} = \mathbf{5.06 \text{ kW}}$$

(b) The adiabatic efficiency of the compressor is determined from

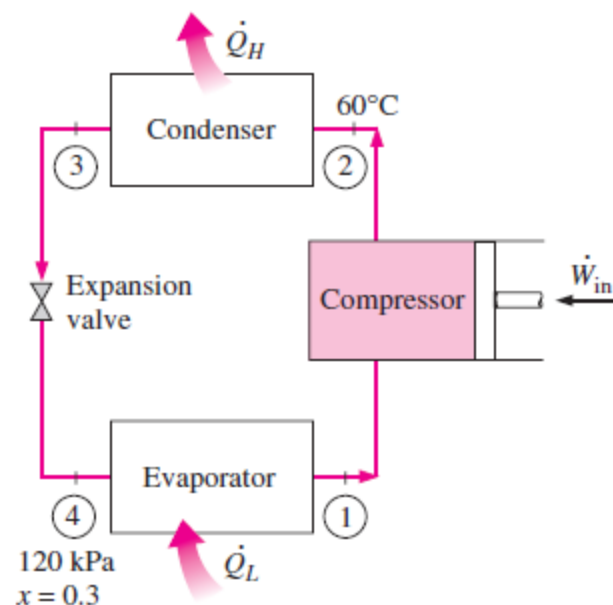
$$\eta_C = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{281.16 - 246.36}{288.53 - 246.36} = \mathbf{82.5\%}$$

(c) The COP of the refrigerator is determined from its definition,

$$\text{COP}_R = \frac{\dot{Q}_L}{\dot{W}_{\text{in}}} = \frac{19.4 \text{ kW}}{5.06 \text{ kW}} = \mathbf{3.83}$$

11-20 Refrigerant-134a enters the compressor of a refrigerator at 140 kPa and -10°C at a rate of $0.3 \text{ m}^3/\text{min}$ and leaves at 1 MPa. The isentropic efficiency of the compressor is 78 percent. The refrigerant enters the throttling valve at 0.95 MPa and 30°C and leaves the evaporator as saturated vapor at -18.5°C . Show the cycle on a T - s diagram with respect to saturation lines, and determine (a) the power input to the compressor, (b) the rate of heat removal from the refrigerated space, and (c) the pressure drop and rate of heat gain in the line between the evaporator and the compressor. *Answers:* (a) 1.88 kW, (b) 4.99 kW, (c) 1.65 kPa, 0.241 kW

11-22 A refrigerator uses refrigerant-134a as the working fluid and operates on the ideal vapor-compression refrigeration cycle. The refrigerant enters the evaporator at 120 kPa with a quality of 30 percent and leaves the compressor at 60°C . If the compressor consumes 450 W of power, determine (a) the mass flow rate of the refrigerant, (b) the condenser pressure, and (c) the COP of the refrigerator. *Answers:* (a) 0.00727 kg/s, (b) 672 kPa, (c) 2.43



Selecting the Right Refrigerant

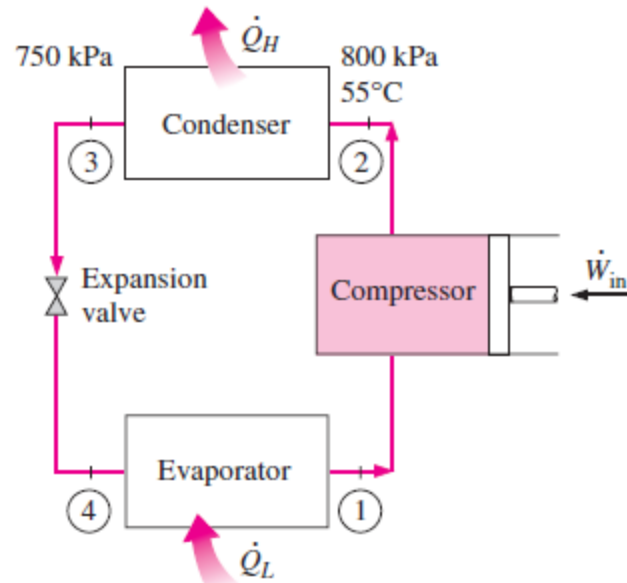
11-26 A refrigerator that operates on the ideal vapor-compression cycle with refrigerant-134a is to maintain the refrigerated space at -10°C while rejecting heat to the environment at 25°C . Select reasonable pressures for the evaporator and the condenser, and explain why you chose those values.

Analysis Allowing a temperature difference of 10°C for effective heat transfer, the evaporation and condensation temperatures of the refrigerant should be -20°C and 35°C , respectively. The saturation pressures corresponding to these temperatures are 0.133 MPa and 0.888 MPa. Therefore, the recommended evaporator and condenser pressures are **0.133 MPa** and **0.888 MPa**, respectively.

11-27 A heat pump that operates on the ideal vapor-compression cycle with refrigerant-134a is used to heat a house and maintain it at 22°C by using underground water at 10°C as the heat source. Select reasonable pressures for the evaporator and the condenser, and explain why you chose those values.

Analysis Allowing a temperature difference of 10°C for effective heat transfer, the evaporation and condensation temperatures of the refrigerant should be 0°C and 32°C , respectively. The saturation pressures corresponding to these temperatures are 0.293 MPa and 0.816 MPa. Therefore, the recommended evaporator and condenser pressures are **0.293 MPa** and **0.816 MPa**, respectively.

11-34 Refrigerant-134a enters the condenser of a residential heat pump at 800 kPa and 55°C at a rate of 0.018 kg/s and leaves at 750 kPa subcooled by 3°C. The refrigerant enters the compressor at 200 kPa superheated by 4°C. Determine (a) the isentropic efficiency of the compressor, (b) the rate of heat supplied to the heated room, and (c) the COP of the heat pump. Also, determine (d) the COP and the rate of heat supplied to the heated room if this heat pump operated on the ideal vapor-compression cycle between the pressure limits of 200 and 800 kPa.



Analysis (a) The properties of refrigerant-134a are (Tables A-11 through A-13)

$$\left. \begin{array}{l} P_2 = 800 \text{ kPa} \\ T_2 = 55^\circ\text{C} \end{array} \right\} h_2 = 291.76 \text{ kJ/kg}$$

$$T_3 = T_{\text{sat}@750 \text{ kPa}} = 29.06^\circ\text{C}$$

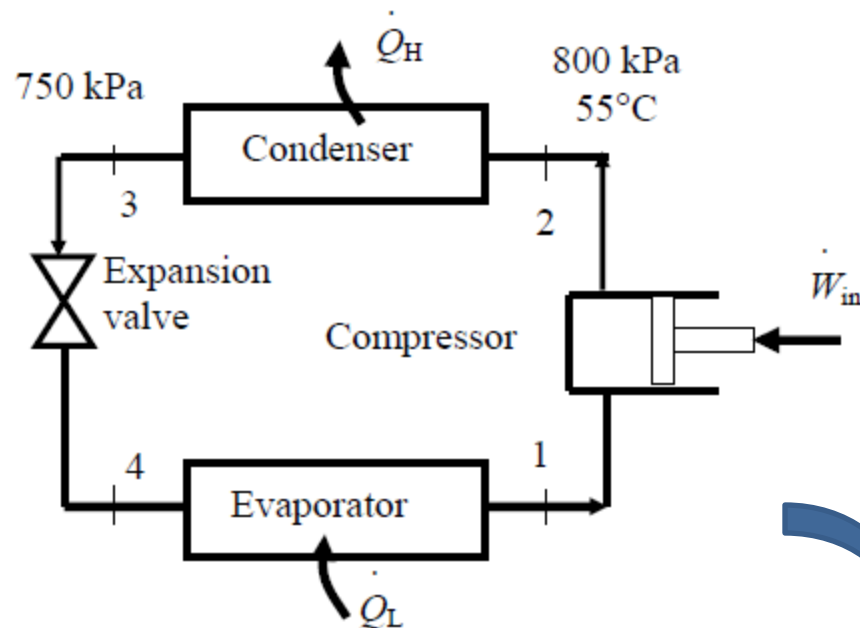
$$\left. \begin{array}{l} P_3 = 750 \text{ kPa} \\ T_3 = (29.06 - 3)^\circ\text{C} \end{array} \right\} h_3 = 87.91 \text{ kJ/kg}$$

$$h_4 = h_3 = 87.91 \text{ kJ/kg}$$

$$T_{\text{sat}@200 \text{ kPa}} = -10.09^\circ\text{C}$$

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa} \\ T_1 = (-10.09 + 4)^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 = 247.87 \text{ kJ/kg} \\ s_1 = 0.9506 \text{ kJ/kg} \end{array}$$

$$\left. \begin{array}{l} P_2 = 800 \text{ kPa} \\ s_2 = s_1 \end{array} \right\} h_{2s} = 277.26$$



The isentropic efficiency of the compressor is

$$\eta_C = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{277.26 - 247.87}{291.76 - 247.87} = \mathbf{0.670}$$

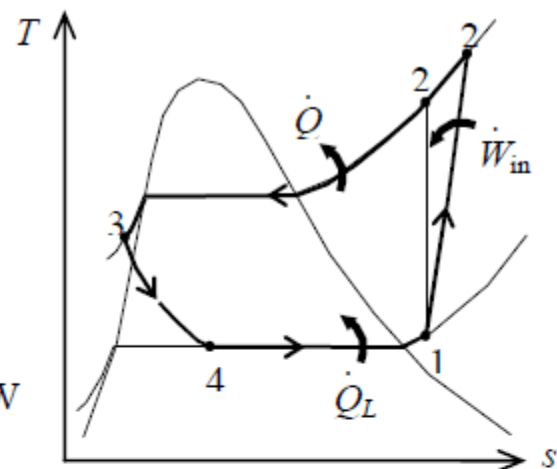
(b) The rate of heat supplied to the room is

$$\dot{Q}_H = \dot{m}(h_2 - h_3) = (0.018 \text{ kg/s})(291.76 - 87.91) \text{ kJ/kg} = \mathbf{3.67 \text{ kW}}$$

(c) The power input and the COP are

$$\dot{W}_{\text{in}} = \dot{m}(h_2 - h_1) = (0.018 \text{ kg/s})(291.76 - 247.87) \text{ kJ/kg} = 0.790 \text{ kW}$$

$$\text{COP} = \frac{\dot{Q}_H}{\dot{W}_{\text{in}}} = \frac{3.67}{0.790} = \mathbf{4.64}$$



(d) The ideal vapor-compression cycle analysis of the cycle is as follows:

$$h_1 = h_{g@200 \text{ kPa}} = 244.46 \text{ kJ/kg}$$

$$s_1 = s_{g@200 \text{ kPa}} = 0.9377 \text{ kJ/kg.K}$$

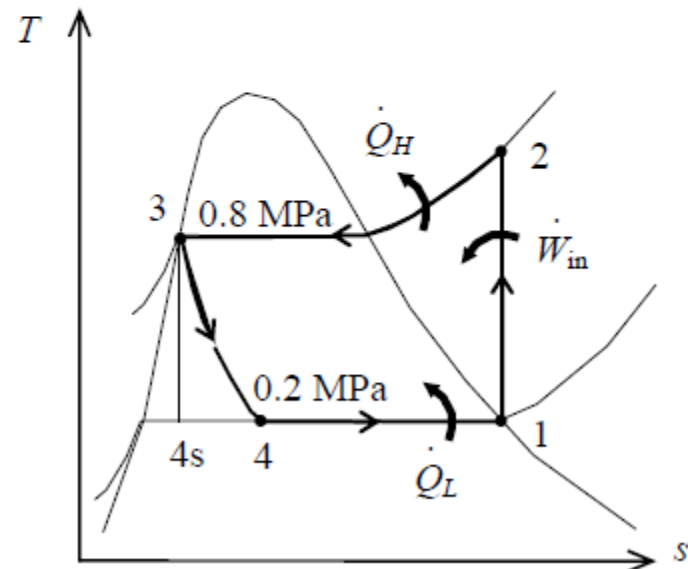
$$\left. \begin{array}{l} P_2 = 800 \text{ MPa} \\ s_2 = s_1 \end{array} \right\} h_2 = 273.25 \text{ kJ/kg}$$

$$h_3 = h_{f@800 \text{ kPa}} = 95.47 \text{ kJ/kg}$$

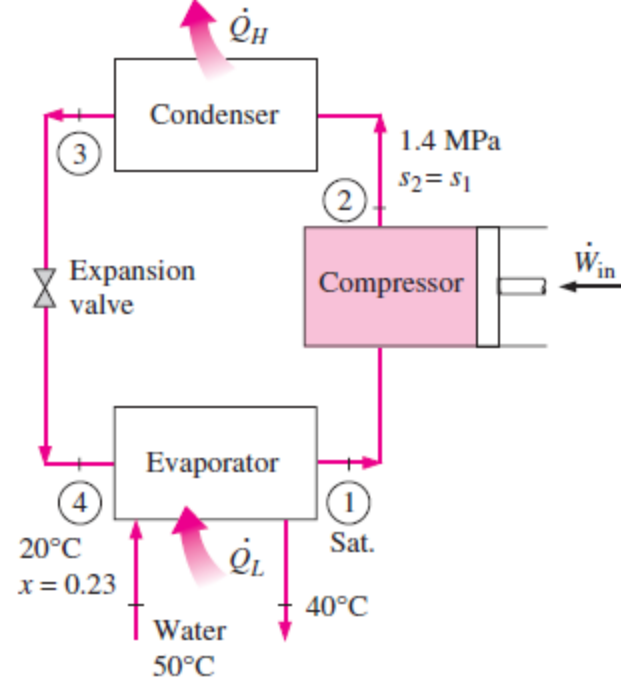
$$h_4 = h_3$$

$$\text{COP} = \frac{h_2 - h_3}{h_2 - h_1} = \frac{273.25 - 95.47}{273.25 - 244.46} = \mathbf{6.18}$$

$$\dot{Q}_H = \dot{m}(h_2 - h_3) = (0.018 \text{ kg/s})(273.25 - 95.47) \text{ kJ/kg} = \mathbf{3.20 \text{ kW}}$$



11-35 A heat pump with refrigerant-134a as the working fluid is used to keep a space at 25°C by absorbing heat from geothermal water that enters the evaporator at 50°C at a rate of 0.065 kg/s and leaves at 40°C. The refrigerant enters the evaporator at 20°C with a quality of 23 percent and leaves at the inlet pressure as saturated vapor. The refrigerant loses 300 W of heat to the surroundings as it flows through the compressor and the refrigerant leaves the compressor at 1.4 MPa at the same entropy as the inlet. Determine (a) the degrees of subcooling of the refrigerant in the condenser, (b) the mass flow rate of the refrigerant, (c) the heating load and the COP of the heat pump, and (d) the theoretical minimum power input to the compressor for the same heating load. *Answers:* (a) 3.8°C, (b) 0.0194 kg/s, (c) 3.07 kW, 4.68, (d) 0.238 kW



11-47 Consider a two-stage cascade refrigeration system operating between the pressure limits of 1.2 MPa and 200 kPa with refrigerant-134a as the working fluid. Heat rejection from the lower cycle to the upper cycle takes place in an adiabatic counterflow heat exchanger where the pressure in the upper and lower cycles are 0.4 and 0.5 MPa, respectively. In both cycles, the refrigerant is a saturated liquid at the condenser exit and a saturated vapor at the compressor inlet, and the isentropic efficiency of the compressor is 80 percent. If the mass flow rate of the refrigerant through the lower cycle is 0.15 kg/s, determine (a) the mass flow rate of the refrigerant through the upper cycle, (b) the rate of heat removal from the refrigerated space, and (c) the COP of this refrigerator.

Answers: (a) 0.212 kg/s, (b) 25.7 kW, (c) 2.68

$$h_1 = h_{g@200 \text{ kPa}} = 244.46 \text{ kJ/kg}$$

$$s_1 = s_{g@200 \text{ kPa}} = 0.9377 \text{ kJ/kg}\cdot\text{K}$$

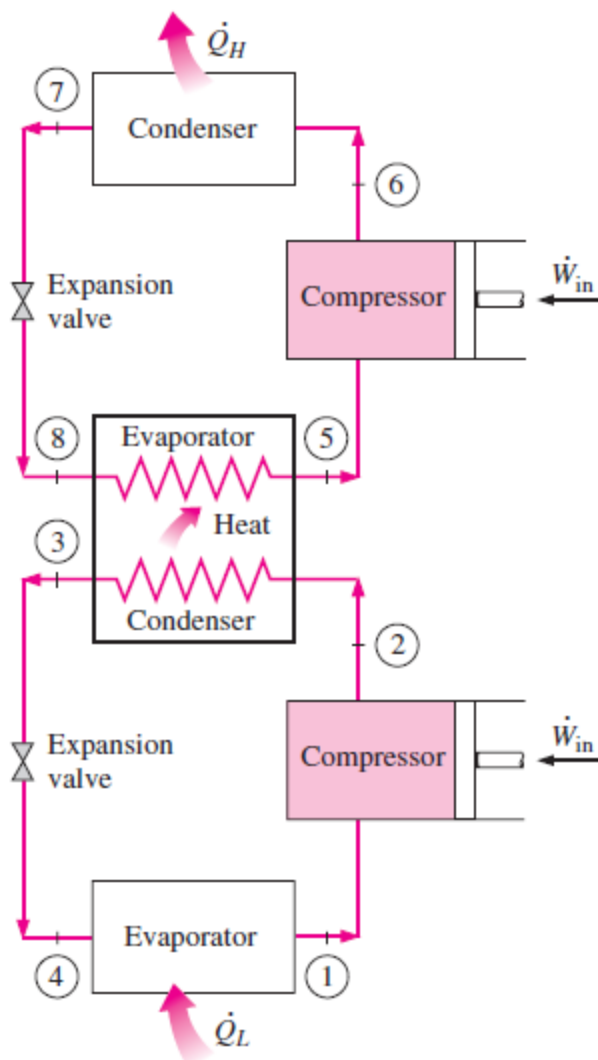
$$\left. \begin{array}{l} P_2 = 500 \text{ kPa} \\ s_2 = s_1 \end{array} \right\} h_{2s} = 263.30 \text{ kJ/kg}$$

$$\eta_C = \frac{h_{2s} - h_1}{h_2 - h_1}$$

$$0.80 = \frac{263.30 - 244.46}{h_2 - 244.46} \longrightarrow h_2 = 268.01 \text{ kJ/kg}$$

$$h_3 = h_{f@500 \text{ kPa}} = 73.33 \text{ kJ/kg}$$

$$h_4 = h_3 = 73.33 \text{ kJ/kg}$$



$$h_5 = h_{g@400 \text{ kPa}} = 255.55 \text{ kJ/kg}$$

$$s_5 = s_{g@400 \text{ kPa}} = 0.9269 \text{ kJ/kg.K}$$

$$\left. \begin{array}{l} P_6 = 1200 \text{ kPa} \\ s_6 = s_5 \end{array} \right\} h_{6s} = 278.33 \text{ kJ/kg}$$

$$\eta_C = \frac{h_{6s} - h_5}{h_6 - h_5}$$

$$0.80 = \frac{278.33 - 255.55}{h_6 - 255.55} \longrightarrow h_6 = 284.02 \text{ kJ/kg}$$

$$h_7 = h_{f@1200 \text{ kPa}} = 117.77 \text{ kJ/kg}$$

$$h_8 = h_7 = 117.77 \text{ kJ/kg}$$

The mass flow rate of the refrigerant through the upper cycle is determined from an energy balance on the heat exchanger

$$\dot{m}_A (h_5 - h_8) = \dot{m}_B (h_2 - h_3)$$

$$\dot{m}_A (255.55 - 117.77) \text{ kJ/kg} = (0.15 \text{ kg/s})(268.01 - 73.33) \text{ kJ/kg} \longrightarrow \dot{m}_A = \mathbf{0.212 \text{ kg/s}}$$

(b) The rate of heat removal from the refrigerated space is

$$\dot{Q}_L = \dot{m}_B (h_1 - h_4) = (0.15 \text{ kg/s})(244.46 - 73.33) \text{ kJ/kg} = \mathbf{25.67 \text{ kW}}$$

(c) The power input and the COP are

$$\begin{aligned} \dot{W}_{\text{in}} &= \dot{m}_A (h_6 - h_5) + \dot{m}_B (h_2 - h_1) \\ &= (0.15 \text{ kg/s})(284.02 - 255.55) \text{ kJ/kg} + (0.212 \text{ kg/s})(268.01 - 244.46) \text{ kJ/kg} = 9.566 \text{ kW} \end{aligned}$$

$$\text{COP} = \frac{\dot{Q}_L}{\dot{W}_{\text{in}}} = \frac{25.67}{9.566} = \mathbf{2.68}$$

Gas Refrigeration Cycle

11-56  Air enters the compressor of an ideal gas refrigeration cycle at 12°C and 50 kPa and the turbine at 47°C and 250 kPa. The mass flow rate of air through the cycle is 0.08 kg/s. Assuming variable specific heats for air, determine (a) the rate of refrigeration, (b) the net power input, and (c) the coefficient of performance.

Answers: (a) 6.67 kW, (b) 3.88 kW, (c) 1.72

Analysis (a) We assume both the turbine and the compressor to be isentropic, the turbine inlet temperature to be the temperature of the surroundings, and the compressor inlet temperature to be the temperature of the refrigerated space. From the air table (Table A-17),

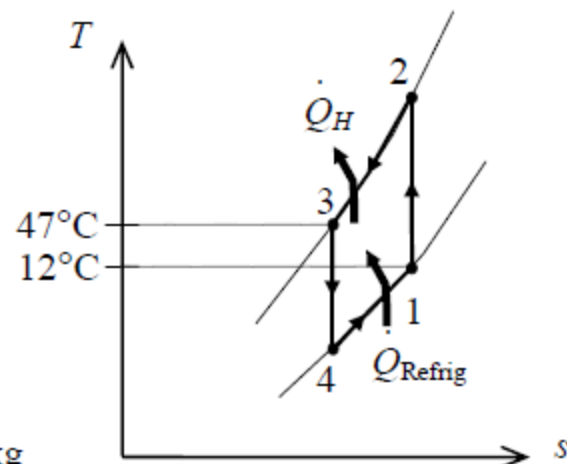
$$T_1 = 285 \text{ K} \longrightarrow h_1 = 285.14 \text{ kJ/kg} \\ P_{r_1} = 1.1584$$

$$T_3 = 320 \text{ K} \longrightarrow h_3 = 320.29 \text{ kJ/kg} \\ P_{r_3} = 1.7375$$

Thus,

$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = \left(\frac{250}{50} \right) (1.1584) = 5.792 \longrightarrow T_2 = 450.4 \text{ K} \\ h_2 = 452.17 \text{ kJ/kg}$$

$$P_{r_4} = \frac{P_4}{P_3} P_{r_3} = \left(\frac{50}{250} \right) (1.7375) = 0.3475 \longrightarrow T_4 = 201.8 \text{ K} \\ h_4 = 201.76 \text{ kJ/kg}$$



Then the rate of refrigeration is

$$\dot{Q}_{\text{refrig}} = \dot{m}(q_L) = \dot{m}(h_1 - h_4) = (0.08 \text{ kg/s})(285.14 - 201.76) \text{ kJ/kg} = \mathbf{6.67 \text{ kW}}$$



(b) The net power input is determined from

$$\dot{W}_{\text{net, in}} = \dot{W}_{\text{comp, in}} - \dot{W}_{\text{turb, out}}$$

where

$$\dot{W}_{\text{comp, in}} = \dot{m}(h_2 - h_1) = (0.08 \text{ kg/s})(452.17 - 285.14) \text{ kJ/kg} = 13.36 \text{ kW}$$

$$\dot{W}_{\text{turb, out}} = \dot{m}(h_3 - h_4) = (0.08 \text{ kg/s})(320.29 - 201.76) \text{ kJ/kg} = 9.48 \text{ kW}$$

Thus,

$$\dot{W}_{\text{net, in}} = 13.36 - 9.48 = \mathbf{3.88 \text{ kW}}$$

(c) The COP of this ideal gas refrigeration cycle is determined from

$$\text{COP}_R = \frac{\dot{Q}_L}{\dot{W}_{\text{net, in}}} = \frac{6.67 \text{ kW}}{3.88 \text{ kW}} = \mathbf{1.72}$$



Repeat Prob. 11-56 for a compressor isentropic efficiency of 80 percent and a turbine isentropic efficiency of 85 percent.

Analysis (a) We assume the turbine inlet temperature to be the temperature of the surroundings, and the compressor inlet temperature to be the temperature of the refrigerated space. From the air table (Table A-17),

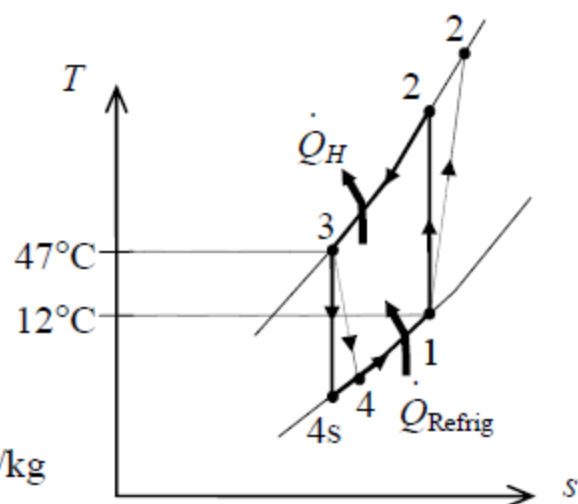
$$T_1 = 285 \text{ K} \longrightarrow \begin{aligned} h_1 &= 285.14 \text{ kJ/kg} \\ P_{r1} &= 1.1584 \end{aligned}$$

$$T_3 = 320 \text{ K} \longrightarrow \begin{aligned} h_3 &= 320.29 \text{ kJ/kg} \\ P_{r3} &= 1.7375 \end{aligned}$$

Thus,

$$P_{r2} = \frac{P_2}{P_1} P_{r1} = \left(\frac{250}{50} \right) (1.1584) = 5.792 \longrightarrow \begin{aligned} T_{2s} &= 450.4 \text{ K} \\ h_{2s} &= 452.17 \text{ kJ/kg} \end{aligned}$$

$$P_{r4} = \frac{P_4}{P_3} P_{r3} = \left(\frac{50}{250} \right) (1.7375) = 0.3475 \longrightarrow \begin{aligned} T_{4s} &= 201.8 \text{ K} \\ h_{4s} &= 201.76 \text{ kJ/kg} \end{aligned}$$



Also,



$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}} \longrightarrow h_4 = h_3 - \eta_T (h_3 - h_{4s})$$

$$= 320.29 - (0.85)(320.29 - 201.76)$$

$$= 219.54 \text{ kJ/kg}$$

Then the rate of refrigeration is

$$\dot{Q}_{\text{refrig}} = \dot{m}(q_L) = \dot{m}(h_1 - h_4) = (0.08 \text{ kg/s})(285.14 - 219.54) \text{ kJ/kg} = \mathbf{5.25 \text{ kW}}$$

(b) The net power input is determined from

$$\dot{W}_{\text{net, in}} = \dot{W}_{\text{comp, in}} - \dot{W}_{\text{turb, out}}$$

where

$$\dot{W}_{\text{comp, in}} = \dot{m}(h_2 - h_1) = \dot{m}(h_{2s} - h_1) / \eta_C$$

$$= (0.08 \text{ kg/s})[(452.17 - 285.14) \text{ kJ/kg}] / (0.80) = 16.70 \text{ kW}$$

$$\dot{W}_{\text{turb, out}} = \dot{m}(h_3 - h_4) = (0.08 \text{ kg/s})(320.29 - 219.54) \text{ kJ/kg} = 8.06 \text{ kW}$$

Thus,

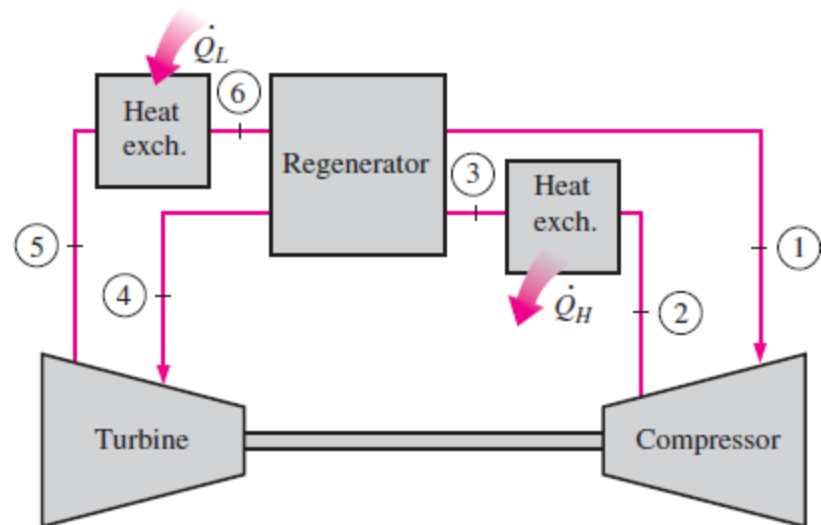
$$\dot{W}_{\text{net, in}} = 16.70 - 8.06 = \mathbf{8.64 \text{ kW}}$$

(c) The COP of this ideal gas refrigeration cycle is determined from

$$\text{COP}_R = \frac{\dot{Q}_L}{\dot{W}_{\text{net, in}}} = \frac{5.25 \text{ kW}}{8.64 \text{ kW}} = \mathbf{0.61}$$

11–60 A gas refrigeration cycle with a pressure ratio of 3 uses helium as the working fluid. The temperature of the helium is -10°C at the compressor inlet and 50°C at the turbine inlet. Assuming adiabatic efficiencies of 80 percent for both the turbine and the compressor, determine (a) the minimum temperature in the cycle, (b) the coefficient of performance, and (c) the mass flow rate of the helium for a refrigeration rate of 18 kW.

11–63 A gas refrigeration system using air as the working fluid has a pressure ratio of 5. Air enters the compressor at 0°C. The high-pressure air is cooled to 35°C by rejecting heat to the surroundings. The refrigerant leaves the turbine at –80°C and then it absorbs heat from the refrigerated space before entering the regenerator. The mass flow rate of air is 0.4 kg/s. Assuming isentropic efficiencies of 80 percent for the compressor and 85 percent for the turbine and using constant specific heats at room temperature, determine (a) the effectiveness of the regenerator, (b) the rate of heat removal from the refrigerated space, and (c) the COP of the cycle. Also, determine (d) the refrigeration load and the COP if this system operated on the simple gas refrigeration cycle. Use the same compressor inlet temperature as given, the same turbine inlet temperature as calculated, and the same compressor and turbine efficiencies. *Answers: (a) 0.434, (b) 21.4 kW, (c) 0.478, (d) 24.7 kW, 0.599*



Analysis (a) From the isentropic relations,

$$T_{2s} = T_1 \left(\frac{P_2}{P_1} \right)^{(k-1)/k} = (273.2 \text{ K})(5)^{0.4/1.4} = 432.4 \text{ K}$$

$$\eta_C = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{T_{2s} - T_1}{T_2 - T_1}$$

$$0.80 = \frac{432.4 - 273.2}{T_2 - 273.2} \longrightarrow T_2 = 472.5 \text{ K}$$

The temperature at state 4 can be determined by solving the following two equations simultaneously:



$$T_{5s} = T_4 \left(\frac{P_5}{P_4} \right)^{(k-1)/k} = T_4 \left(\frac{1}{5} \right)^{0.4/1.4}$$

$$\eta_T = \frac{h_4 - h_5}{h_4 - h_{5s}} \rightarrow 0.85 = \frac{T_4 - 193.2}{T_4 - T_{5s}}$$

Using EES, we obtain $T_4 = 281.3$ K.

An energy balance on the regenerator may be written as

$$\dot{m} c_p (T_3 - T_4) = \dot{m} c_p (T_1 - T_6) \longrightarrow T_3 - T_4 = T_1 - T_6$$

or,

$$T_6 = T_1 - T_3 + T_4 = 273.2 - 308.2 + 281.3 = 246.3 \text{ K}$$

The effectiveness of the regenerator is

$$\varepsilon_{\text{regen}} = \frac{h_3 - h_4}{h_3 - h_6} = \frac{T_3 - T_4}{T_3 - T_6} = \frac{308.2 - 281.3}{308.2 - 246.3} = \mathbf{0.434}$$

(b) The refrigeration load is

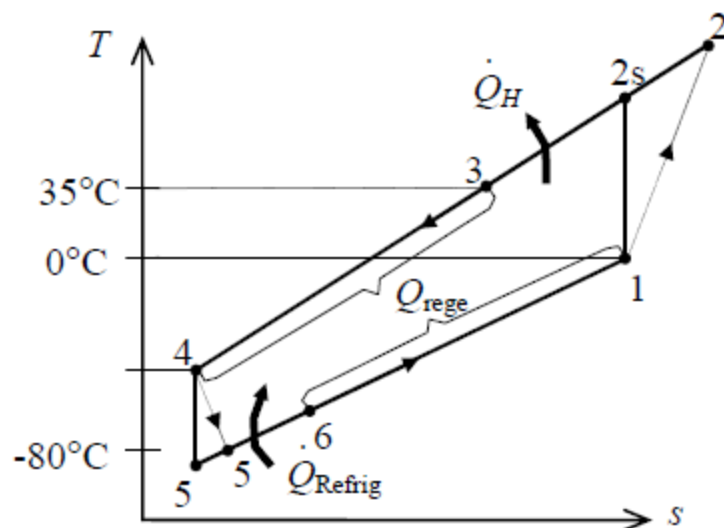
$$\dot{Q}_L = \dot{m} c_p (T_6 - T_5) = (0.4 \text{ kg/s})(1.005 \text{ kJ/kg.K})(246.3 - 193.2) \text{ K} = \mathbf{21.36 \text{ kW}}$$

(c) The turbine and compressor powers and the COP of the cycle are

$$\dot{W}_{C,\text{in}} = \dot{m} c_p (T_2 - T_1) = (0.4 \text{ kg/s})(1.005 \text{ kJ/kg.K})(472.5 - 273.2) \text{ K} = 80.13 \text{ kW}$$

$$\dot{W}_{T,\text{out}} = \dot{m} c_p (T_4 - T_5) = (0.4 \text{ kg/s})(1.005 \text{ kJ/kg.K})(281.3 - 193.2) \text{ kJ/kg} = 35.43 \text{ kW}$$

$$\text{COP} = \frac{\dot{Q}_L}{\dot{W}_{\text{net,in}}} = \frac{\dot{Q}_L}{\dot{W}_{C,\text{in}} - \dot{W}_{T,\text{out}}} = \frac{21.36}{80.13 - 35.43} = \mathbf{0.478}$$



Absorption Refrigeration Systems

11-70 An absorption refrigeration system that receives heat from a source at 130°C and maintains the refrigerated space at -5°C is claimed to have a COP of 2. If the environment temperature is 27°C , can this claim be valid? Justify your answer.

11-71 An absorption refrigeration system receives heat from a source at 120°C and maintains the refrigerated space at 0°C . If the temperature of the environment is 25°C , what is the maximum COP this absorption refrigeration system can have?

11-72 Heat is supplied to an absorption refrigeration system from a geothermal well at 130°C at a rate of $5 \times 10^5 \text{ kJ/h}$. The environment is at 25°C , and the refrigerated space is maintained at -30°C . Determine the maximum rate at which this system can remove heat from the refrigerated space.

Answer: $5.75 \times 10^5 \text{ kJ/h}$

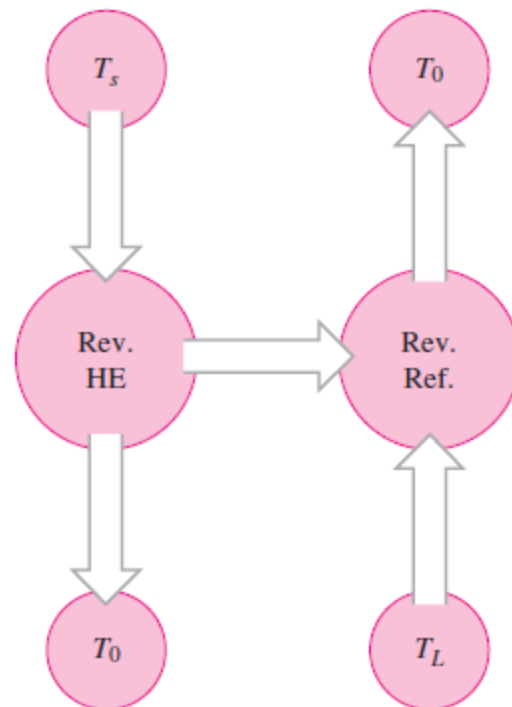
Analysis The maximum COP that this refrigeration system can have is

$$\text{COP}_{\text{R,max}} = \left(1 - \frac{T_0}{T_s}\right) \left(\frac{T_L}{T_0 - T_L}\right) = \left(1 - \frac{298 \text{ K}}{403 \text{ K}}\right) \left(\frac{243}{298 - 243}\right) = 1.15$$

Thus,

$$\dot{Q}_{\text{L,max}} = \text{COP}_{\text{R,max}} \dot{Q}_{\text{gen}} = (1.15)(5 \times 10^5 \text{ kJ/h}) = \mathbf{5.75 \times 10^5 \text{ kJ/h}}$$

11-74 A reversible absorption refrigerator consists of a reversible heat engine and a reversible refrigerator. The system removes heat from a cooled space at -10°C at a rate of 22 kW. The refrigerator operates in an environment at 25°C . If the heat is supplied to the cycle by condensing saturated steam at 200°C , determine (a) the rate at which the steam condenses and (b) the power input to the reversible refrigerator. (c) If the COP of an actual absorption chiller at the same temperature limits has a COP of 0.7, determine the second law efficiency of this chiller. *Answers: (a) 0.00408 kg/s, (b) 2.93 kW, (c) 0.252*



Analysis (a) The thermal efficiency of the reversible heat engine is

$$\eta_{\text{th,rev}} = 1 - \frac{T_0}{T_s} = 1 - \frac{(25 + 273.15) \text{ K}}{(200 + 273.15) \text{ K}} = 0.370$$

The COP of the reversible refrigerator is

$$\text{COP}_{\text{R,rev}} = \frac{T_L}{T_0 - T_L} = \frac{(-10 + 273.15) \text{ K}}{(25 + 273.15) - (-10 + 273.15) \text{ K}} = 7.52$$

The COP of the reversible absorption refrigerator is

$$\text{COP}_{\text{abs,rev}} = \eta_{\text{th,rev}} \text{COP}_{\text{R,rev}} = (0.370)(7.52) = 2.78$$

The heat input to the reversible heat engine is

$$\dot{Q}_{\text{in}} = \frac{\dot{Q}_L}{\text{COP}_{\text{abs,rev}}} = \frac{22 \text{ kW}}{2.78} = 7.911 \text{ kW}$$



Then, the rate at which the steam condenses becomes

$$\dot{m}_s = \frac{\dot{Q}_{\text{in}}}{h_{fg}} = \frac{7.911 \text{ kJ/s}}{1939.8 \text{ kJ/kg}} = \mathbf{0.00408 \text{ kg/s}}$$

(b) The power input to the refrigerator is equal to the power output from the heat engine

$$\dot{W}_{\text{in,R}} = \dot{W}_{\text{out,HE}} = \eta_{\text{th,rev}} \dot{Q}_{\text{in}} = (0.370)(7.911 \text{ kW}) = \mathbf{2.93 \text{ kW}}$$

(c) The second-law efficiency of an actual absorption chiller with a COP of 0.7 is

$$\eta_{\text{II}} = \frac{\text{COP}_{\text{actual}}}{\text{COP}_{\text{abs,rev}}} = \frac{0.7}{2.78} = \mathbf{0.252}$$

Special Topic: Thermoelectric Power Generation and Refrigeration Systems

11-84 A thermoelectric refrigerator removes heat from a refrigerated space at -5°C at a rate of 130 W and rejects it to an environment at 20°C . Determine the maximum coefficient of performance this thermoelectric refrigerator can have and the minimum required power input. *Answers: 10.72, 12.1 W*

11-85 A thermoelectric cooler has a COP of 0.15 and removes heat from a refrigerated space at a rate of 180 W. Determine the required power input to the thermoelectric cooler, in W.

11–87 A thermoelectric refrigerator is powered by a 12-V car battery that draws 3 A of current when running. The refrigerator resembles a small ice chest and is claimed to cool nine canned drinks, 0.350-L each, from 25 to 3°C in 12 h. Determine the average COP of this refrigerator.



Properties The properties of canned drinks are the same as those of water at room temperature, $\rho = 1 \text{ kg/L}$ and $c_p = 4.18 \text{ kJ/kg}\cdot^\circ\text{C}$ (Table A-3).

Analysis The cooling rate of the refrigerator is simply the rate of decrease of the energy of the canned drinks,

$$m = \rho V = 9 \times (1 \text{ kg/L})(0.350 \text{ L}) = 3.15 \text{ kg}$$

$$Q_{\text{cooling}} = mc\Delta T = (3.15 \text{ kg})(4.18 \text{ kJ/kg}\cdot^\circ\text{C})(25 - 3)^\circ\text{C} = 290 \text{ kJ}$$

$$\dot{Q}_{\text{cooling}} = \frac{Q_{\text{cooling}}}{\Delta t} = \frac{290 \text{ kJ}}{12 \times 3600 \text{ s}} = 0.00671 \text{ kW} = 6.71 \text{ W}$$

The electric power consumed by the refrigerator is

$$\dot{W}_{\text{in}} = VI = (12 \text{ V})(3 \text{ A}) = 36 \text{ W}$$

Then the COP of the refrigerator becomes

$$\text{COP} = \frac{\dot{Q}_{\text{cooling}}}{\dot{W}_{\text{in}}} = \frac{6.71 \text{ W}}{36 \text{ W}} = \mathbf{0.186} \approx 0.20$$

11–89 It is proposed to run a thermoelectric generator in conjunction with a solar pond that can supply heat at a rate of 10^6 kJ/h at 80°C . The waste heat is to be rejected to the environment at 30°C . What is the maximum power this thermoelectric generator can produce?

Chapter 11

End of the Problems